CY4001: Industrial Chemistry

Unit 4:Polymer Chemistry
Lecture 4 : Polymer Revolution

Unit 4:Polymer Chemistry

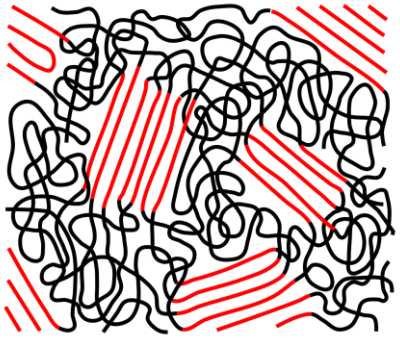
Plastics

Polymers are long chains of molecules called monomers, while plastics are a type of polymer that can be molded into shapes. Plastics are 'synthetic organic polymers', that can be moulded in any form with the help of heat and pressure.

(a) Thermoplastics

- They soften on heating and harden on cooling. Resoftening and rehardening is also possible.
- They consist of non-crosslinked polymers, often with a semi-crystalline structure and glass transition temperature

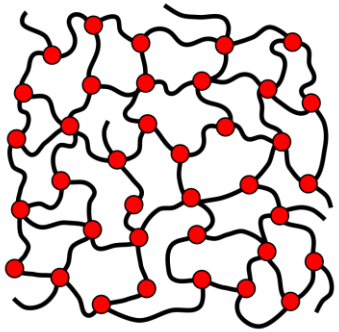
Examples: Polythene, PVC, Polystyrene, Polyester and Polypropylene



(b) Thermosetting

- They set on heating and cannot be resoftened on cooling.
- They consist of closely cross-linked polymers.

Examples: Urea-Formaldehyde, Phenol-Formaldehyde, Alkyl resin, Silicon resin



S.No	Thermoplastics	Thermosetting Plastics
1.	They are prepared by addition polymerization method.	They are prepared by condensation polymerization method.
2.	They are generally linear polymers.	They are generally cross-linked polymers.
3.	They are softer, weaker and less brittle.	They are harder, stronger and more brittle.
4.	They are polymers of smaller molecular weight.	They are polymers of higher molecular weight.
5.	They have low thermal stability.	They have relatively high thermal stability, but prolong heating result in charring.
6.	They can be remoulded.	They cannot be remoulded.
7.	The process is reversible	The process is irreversible.
8.	They soften on heating and regain their original properties on cooling.	They set on heating and cannot be resoftened on cooling.
9.	They are easily broken because the molecules are held together by weak van derWaals forces.	They are not easily broken because the molecules are held together by strong covalent bonds.
10.	They expand very much on heating.	They expand in a limited way.
11.	Hardening is not due to chemical reaction.	Hardening is due to chemical reaction.
12.	Scraps can be reused.	Scraps cannot be reused.
13.	They are moulded by injection moulding process.	They are moulded by compression moulding process.
14.	They are soluble in organic solvents.	They are insoluble in organic solvents.
15.	**Example:** Polyethylene, Polypropylene, PVC, etc.	**Example:** Bakelite, Urea-formaldehyde, etc.

Unit 4:Polymer Chemistry

Vinyl Polymers

Vinyl polymers are a group of polymers derived from substituted vinyl ($H_2C{=}CHR$) monomers. Their backbone is an extended alkane chain [$-CH_2-CHR-$]

Polyethylene also referred to as polythene

- Polyethylene was first synthesized by the German chemist Hans von Pechmann, who prepared it by accident in 1898 while investigating diazomethane. They later characterized the white, waxy substance and recognized that it contained long −CH2− chains and termed it polymethylene.

1) Low Density PE

- Ductile;
- Fit to make shopping bags, plastic bags, food containers and all sorts of disposable packaging.

2) Medium Density PE

- Great density;
- Fit to make gas pipes, shrink wrap, carrier bags, etc.

3) High Density PE

- More rigid;
- Fit to make plastic bottles, water pipes, laundry detergent bottles, etc.

4) Ultra-high Density PE

- High density & abrasion resistant;
- Fit to make biomaterial for hip, knee, and spine implants, etc.

Unit 4:Polymer Chemistry

Vinyl Polymers Polyethylene

Properties

•**Thermoplastic**
Remoulded by heating

•**Durable**:
Resistant to fading, chipping

•**Weatherproof**: outdoor use

•**Low cost**:
Most affordable plastics

•**Non-conductive**:
Doesn't conduct electricity

Applications

•**Packaging**

•**Building construction**

•**Electrical insulation**

•**Submarine cables**

•**Frozen food bags**

•**Cereal liners**

Unit 4:Polymer Chemistry

Vinyl Polymers

Polypropylene

	Polyethylene (PE)	**Polypropylene (PP)**
Flexibility	More flexible	Less flexible
Heat resistance	Less heat resistant	More heat resistant
Common uses	Plastic bags, films, and packaging	Automotive parts, rigid containers, and packaging
Other properties	Weatherproof, good low temperature toughness	Tough, good fatigue resistance, easy to color and mold

Unit 4:Polymer Chemistry

Which One Should I Pick for My Packaging Solution?

The numbers 1–7 on plastics are resin identification codes, indicate the type of plastic

PS: Polystyrene PE: Polyethylene PP: Polypropylene PVC: Polyvinyl Chloride
PET (Polyethylene Terephthalate)

Unit 4:Polymer Chemistry

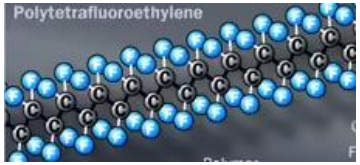
Which One Should I Pick for My Packaging Solution?

PS (Polystyrene)	PP (Polypropylene)	PVC (Polyvinyl Chloride)	PET (Polyethylene Terephthalate)
Translucent	Transparent	A light blue tint	Transparent
Solid plastic form or foam form	Stable, flexible, tough & durable	Durable & great tensile strength	Lightweight yet high-strength barrier
Food-grade plastic material	Fit to make laboratory equipment, medical devices, plastic straws, hot food containers, etc	Not safe for storing food items	Safe for packaging food and drinks

Unit 4:Polymer Chemistry

Teflon (PTFE)

Polytetrafluoroethylene (PTFE) is a synthetic, non-stick, fluoropolymer plastic. It's also known by the brand name Teflon.

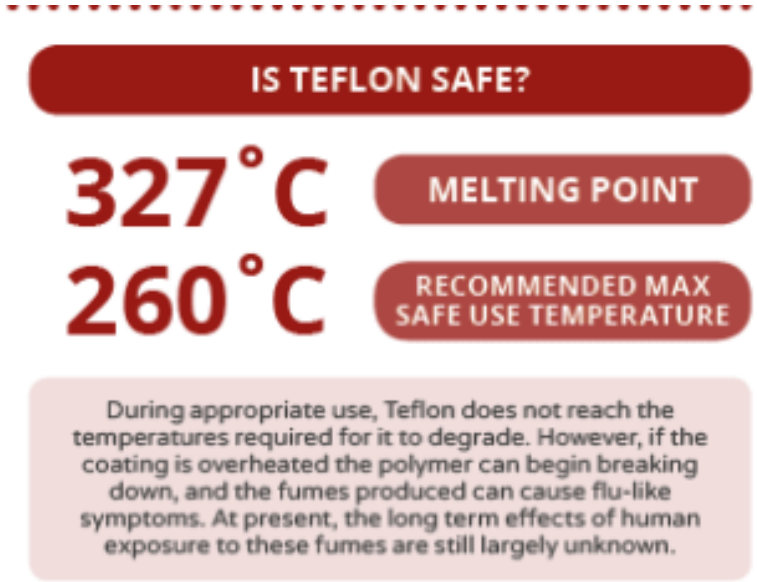


Preparation

Dr. Roy J. Plunkett accidentally discovered Teflon in 1938 while working on refrigerants at the DuPont Company.

$$n\,(CF_2{=}CF_2) \xrightarrow{\text{Polymerization}} -(CF_2-CF_2)_n-$$

Tetrafluoroethylene PTFE

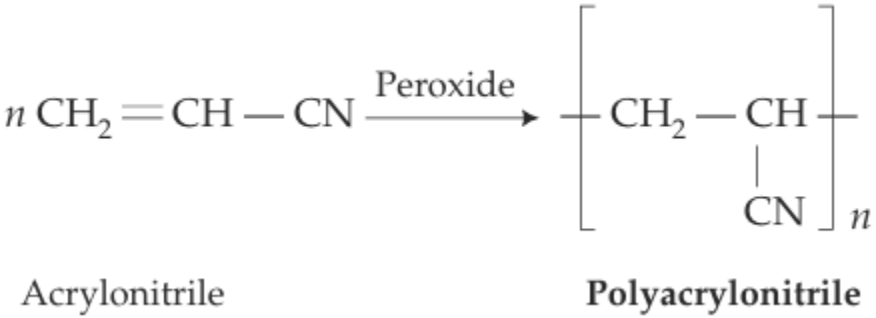


Teflon nonstick properties are in part due to the strength the C-F bond which is chemically inert. PTFE is surround by fluorine atoms and they repel almost all other materia preventing them from adhering to Teflon.

Unit 4:Polymer Chemistry

Orlon

Polyacrylonitrile (PAN) is a polymer that is also known by the brand name Orlon. It is a synthetic acrylic fiber that is used to make clothing, outdoor products. It is used as a substitute for wool in making fi bres.



Properties

- Orlon is resistant to wrinkles, chemicals, UV light, weathering, insects, mildew, and moisture
- It's also resistant to sunlight and greenhouse gases
- It is hypoallergenic and suitable for individuals with wool allergies

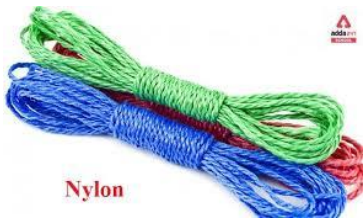
Is orlon toxic?

Orlon can be toxic if burned. When burned, Orlon can produce hydrogen cyanide, a highly toxic gas.

Unit 4:Polymer Chemistry

Nylon

Nylon is a generic name for a family of synthetic polymers called polyamides. These polymers are made up of structural units linked by amide functional groups.



Wallace Carothers synthesized the first nylon 66 in 1936

$$n\ H_2N{-}(CH_2)_6{-}NH_2 + n\ HOOC{-}(CH_2)_4{-}COOH{-}$$

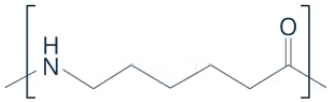
Hexamethylenediamine Adipic acid

$$\downarrow$$

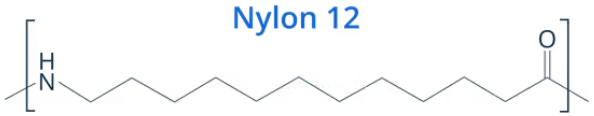
$$-[HN{-}(CH_2)_6{-}NH{-}CO{-}(CH_2)_4{-}CO]_n + 2n\ H_2O$$

Nylon-6,6

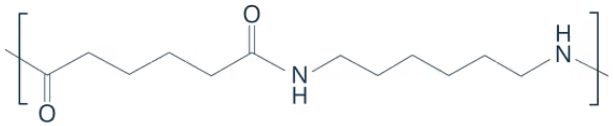
Nylon 6



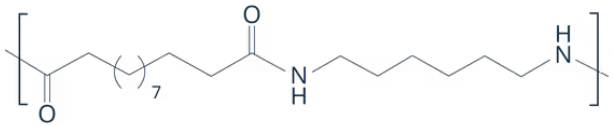
Nylon 12



Nylon 6,6



Nylon 6,12



It is used for making nylon thread in textile materials. • Nylons are used for making fi laments for ropes, bristles for toothbrushes and fi lms etc.,

Polyethylene terephthalate (PET)

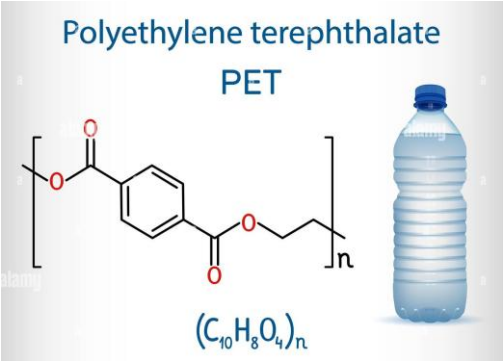
1978: Coca-Cola and Pepsi, introduced the first 2l PET plastic bottles to the World.

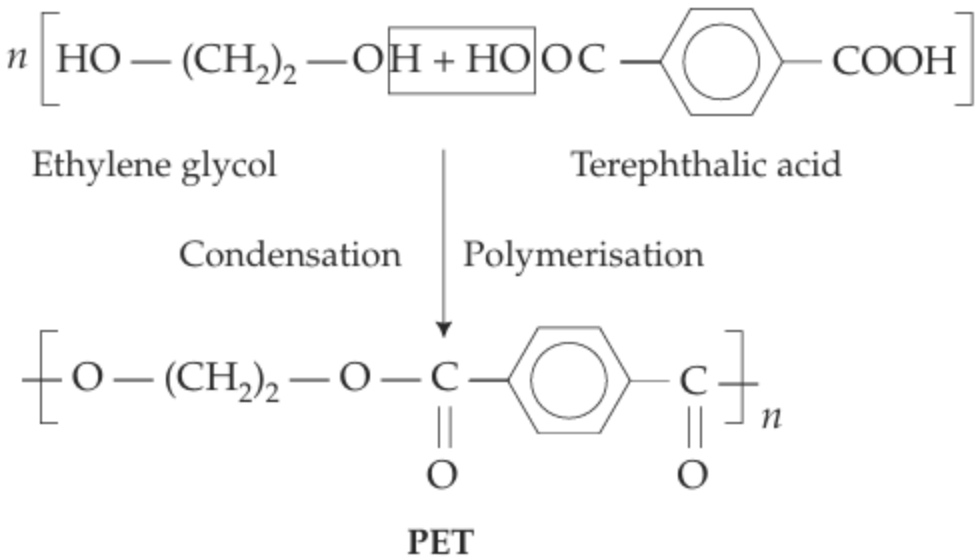
Since 1978, humans contributed unconsciously to spread these plastic bottles over the oceans, beaches, cities and lands.

Unit 4:Polymer Chemistry

Polyethylene terephthalate (PET)

Polyethylene terephthalate (PET) is a thermoplastic polymer that is used in many industries, including packaging, textiles, and engineering





It is a good fi bre-forming material with high tensile strength and can be converted into commercial fi bres. • It possesses high stretch-resistance, high-crease and wrinkle resistance.

Unit 4:Polymer Chemistry

Self-healing Polymers

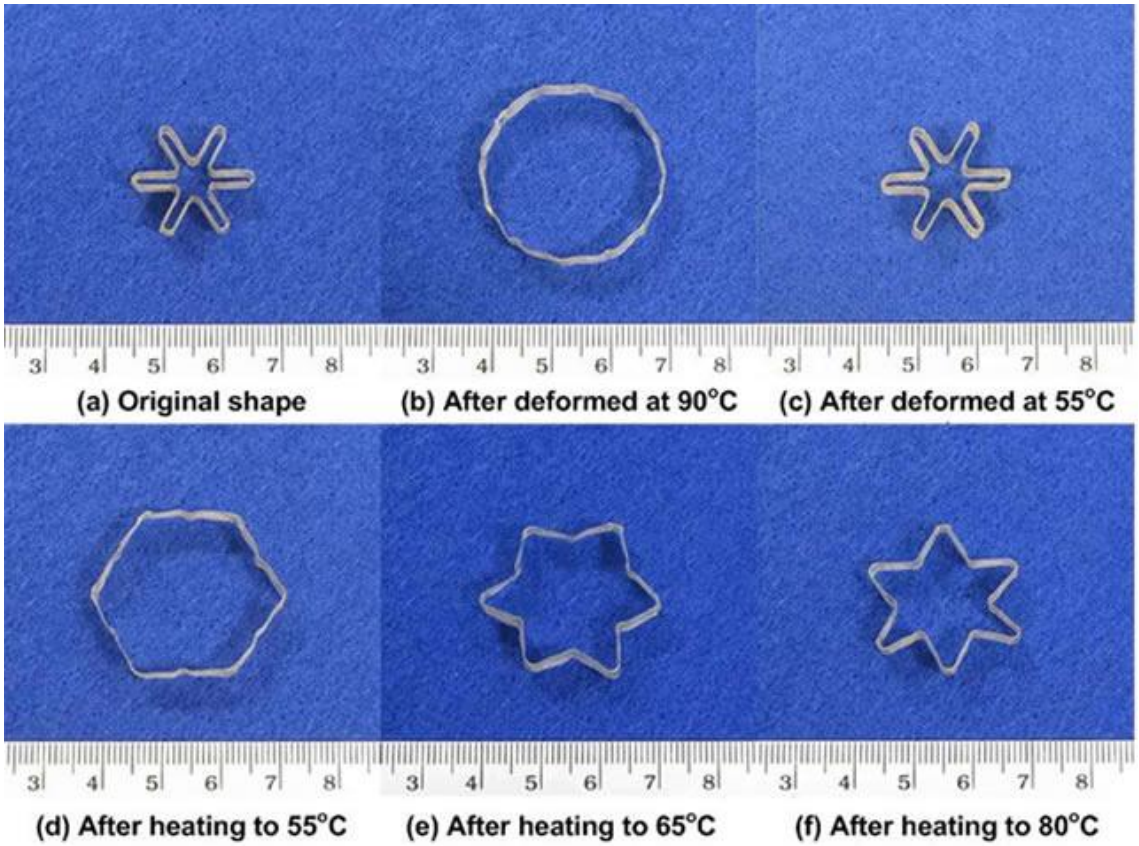
Researchers have developed a stronger than bone and self-healing material

Nature Reviews Materials **volume 5**, pages562–583 (2020)

Unit 4:Polymer Chemistry

Smart Polymers

Scientists have developed a class of polymers allowing it to change its shape, permeability or conductivity when a stimulus is applied to it. The stimulus can be a change in temperature, pH, light intensity or can be of electrical, magnetic or chemical nature.



Unit 4:Polymer Chemistry

How do we engineer the properties of polymers?

Monomers

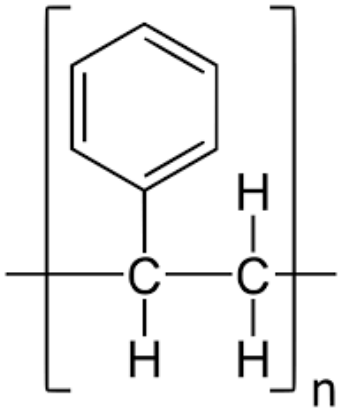
- Monomers decide the nature of the polymer

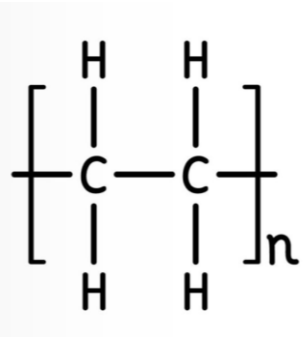
Molecular weight – degree of polymerization

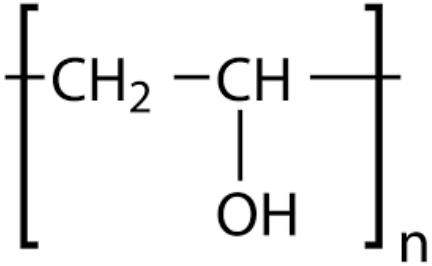
- The distribution of long chain and short chain polymers
- Longer the chain length, stronger will be the polymer

Interactions – Intermolecular forces

The three main types of intermolecular forces in polymers are dispersion forces, dipole-dipole interactions, and hydrogen bonding.





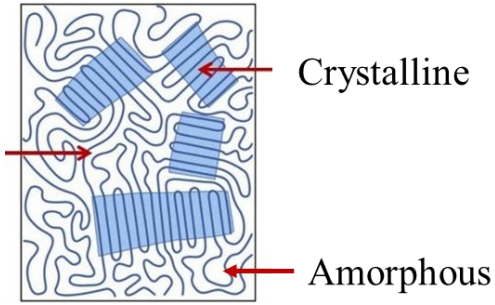


Unit 4:Polymer Chemistry

How do we engineer the properties of polymers?

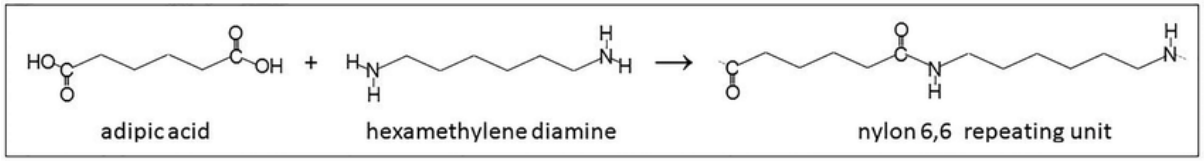
Density/Crystallinity

- Processing
- Branching
- Tacticity

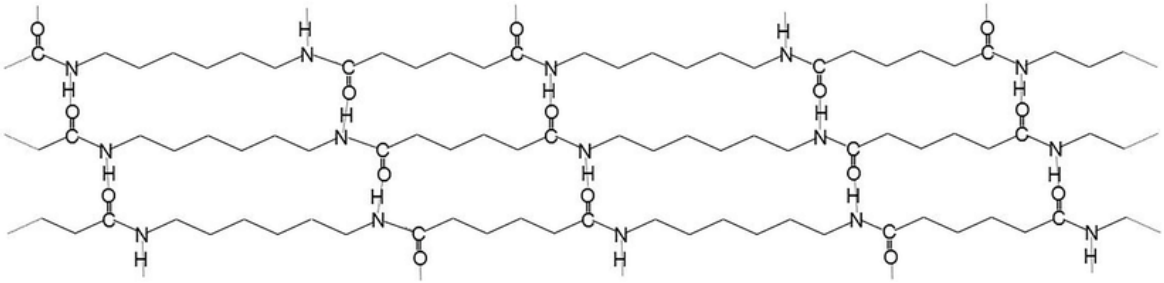


Cross-linking

- Chemical cross-linking can make a material more rigid and give it a better-defined shape.



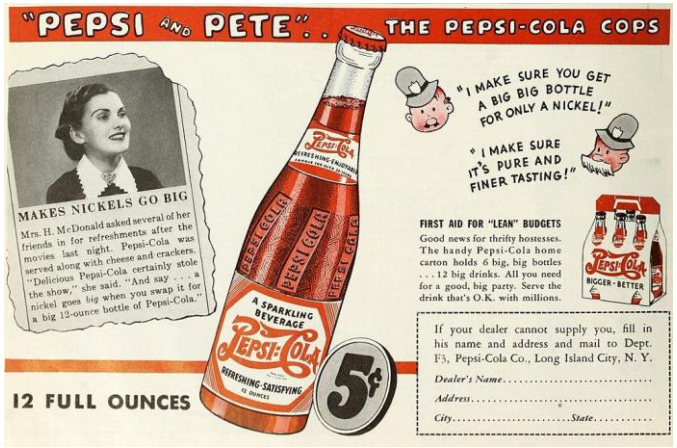
- Cross-linking can also be used to create functional materials like thermosets, rubbers, and hydrogels



Cross-linking can be a ionic or covalent or hydrogen bonds

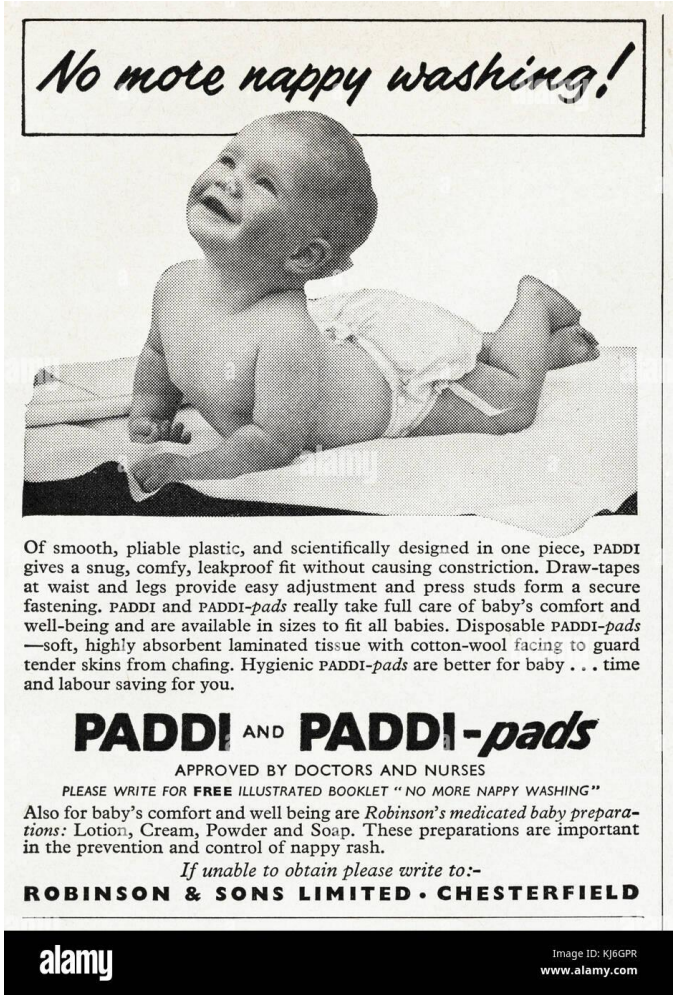
The Polymer Revolution

Polymer has changed our lifestyle since its inception. It has made our lives easier, cheaper and quite colorful.



Bottles lighter than glass, but stronger

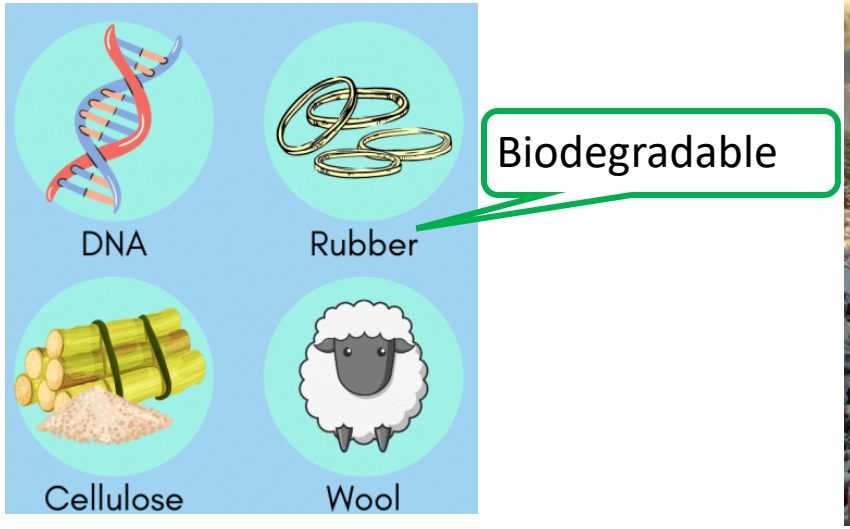
Material similar to silk but much more durable and cheaper



Why it is important

World's best polymer engineer

Nature-made Polymers



Man-made Polymers

- We buy one million plastic bottles/minute (globally) and 91% of them are not recycled.
- Only 7% of all plastics are recycled
- Some polymers are toxic and carcinogenic (polystyrene) and thus poisoning the sea-food.

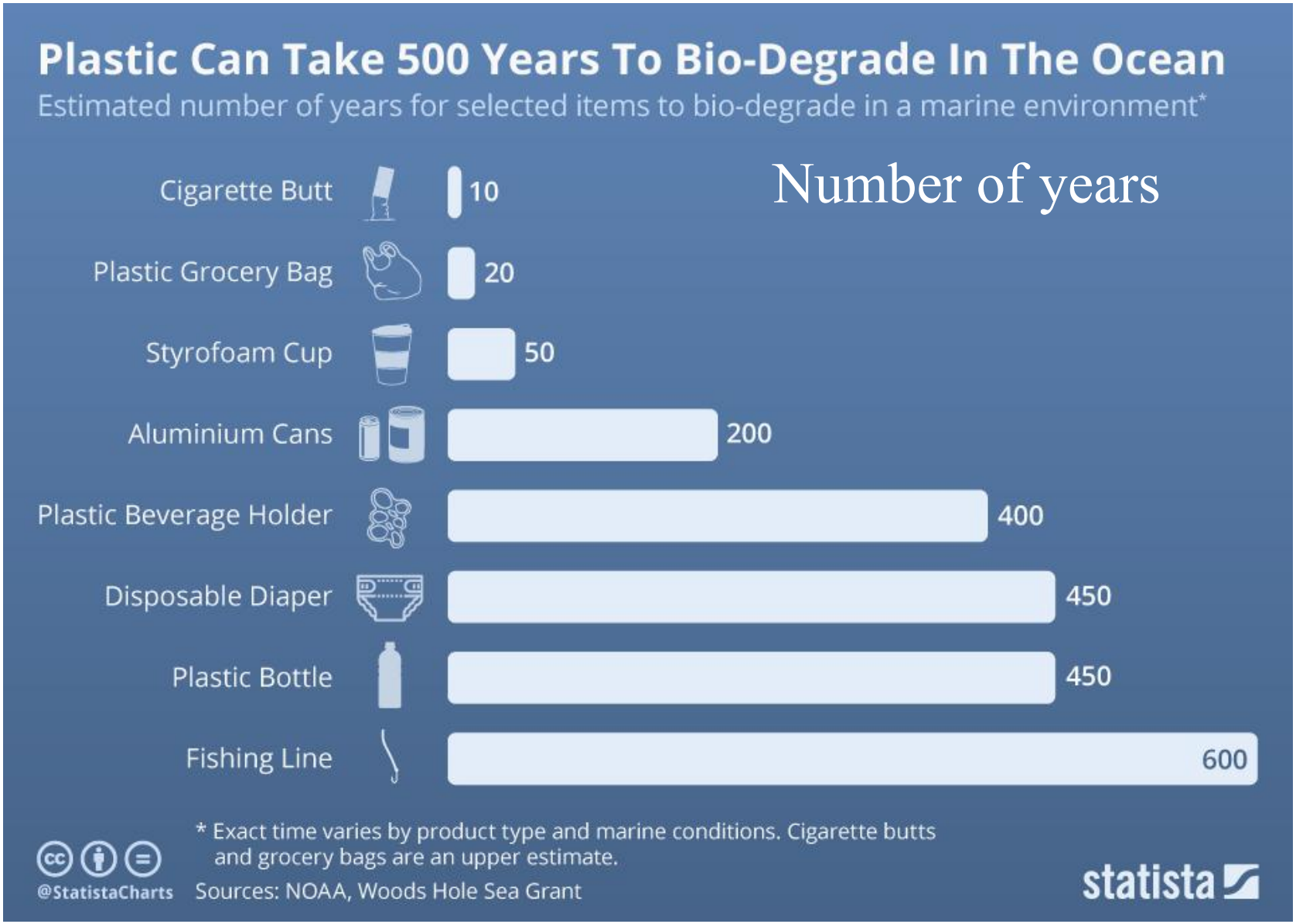
Plastics in Ocean

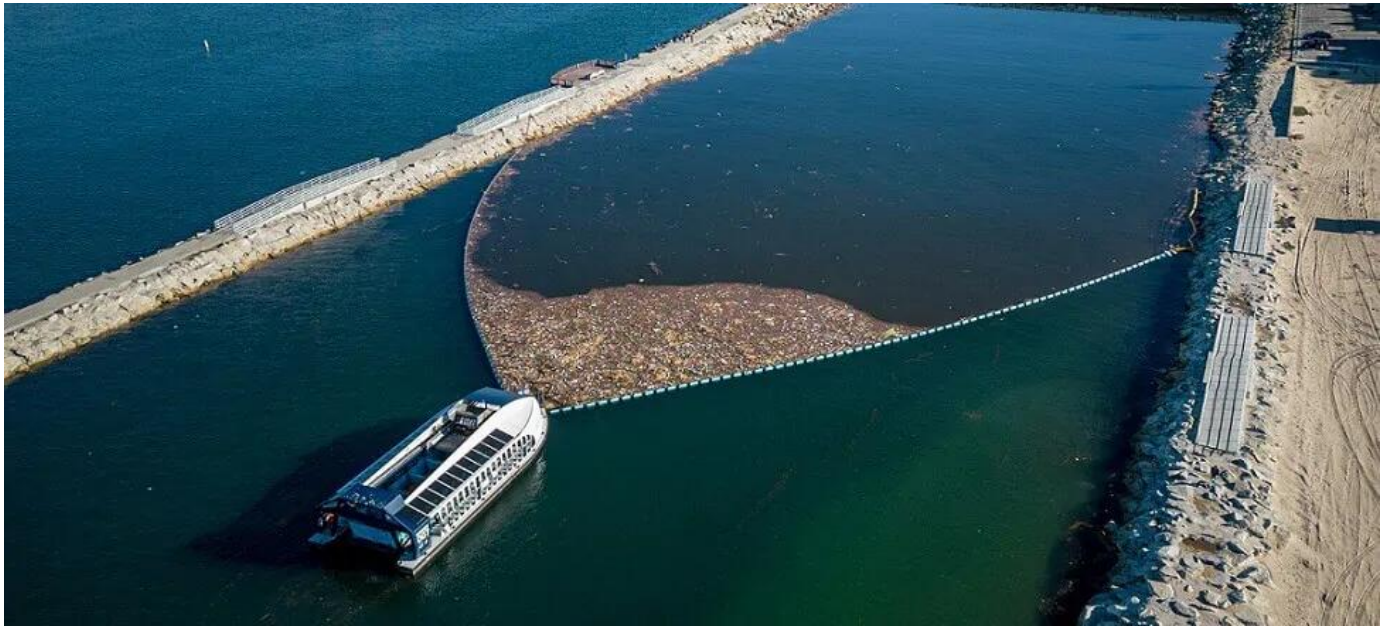
- Plastic pollution kills an estimated 100,000 marine animals each year. 81 out of 123 marine mammal species have been found entangled in plastic, and all seven sea turtle species are affected.

- In 2050, the plastic in the ocean = weight of all fish

We are using things in the minute time-scale, that takes hundreds of years to decompose in the ocean

Estimated number of years to bio-degrade





- ✓ Durable polymers
- ✓ Cleaning the ocean
- ✓ Biodegradable